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RESEARCH PROFILE

- Specializing in AI-driven cybersecurity and adaptive Intrusion Detection Systems for Industrial IoT networks.
- Seeking a postdoctoral position to leverage expertise in deep learning and Game-Theoretic frameworks to develop robust, resource-optimized security frameworks for next-generation networks.

AREAS OF INTEREST

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|---------------------------|----------------------|--------------------|
| ➤ Artificial Intelligence | ➤ Deep Learning | ➤ Machine Learning |
| ➤ Intrusion Detection | ➤ Internet of Things | ➤ Game Theory |

EDUCATION

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|----------------|--|
| 2021 – Present | ➤ Doctor of Philosophy (Ph.D.) — Computer Science and Engineering.
1. Indian Institute of Technology Guwahati, Assam, India, <i>jointly with</i>
2. Indian Institute of Technology (BHU) Varanasi, Uttar Pradesh, India.
CGPA: 8.75/10.00
Thesis title: <i>Resource-Optimized Mimicry Detection in Industrial Internet of Things Networks Using Deep Learning Techniques.</i> |
| 2019 – 2021 | ➤ Master of Technology (M.Tech.) — Computer Science and Engineering.
National Institute of Technology Durgapur (NIT Durgapur), West Bengal, India.
CGPA: 9.28/10.00
Thesis title: <i>A Game Theoretic Framework for Surplus Food Distribution in Smart Cities and Beyond.</i> |
| 2006 – 2010 | ➤ Bachelor of Engineering (B.E.) — Computer Science and Engineering.
University Institute of Technology (UIT), West Bengal, India.
Affiliated with The University of Burdwan (BU).
Result: 79.9% |
| 2006 | ➤ CBSE AISSCE (Class XII).
Kendriya Vidyalaya Number 1, Saltlake, Kolkata, West Bengal, India.
Subjects: Physics, Chemistry, Mathematics, Biology, English.
Result: 79.6% |
| 2004 | ➤ CBSE AISSE (Class X).
Kendriya Vidyalaya Number 1, Saltlake, Kolkata, West Bengal, India.
Result: 82.8% |

RESEARCH EXPERIENCE

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| 2021 – Present | ➤ Doctor of Philosophy (Ph.D.): IIT Guwahati and IIT (BHU) Varanasi
Focus: AI-Driven Cybersecurity & Adaptive IDS for IoT Networks
➤ Developed ensemble frameworks, using Graph Convolutional Networks (GCNs) and GRUs, leveraging spatio-temporal graph diffusion for explainable, zero-day mimicry attack detection at the IoT edge. |
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RESEARCH EXPERIENCE (continued)

- Engineered conjoint detection mechanisms, using count-min sketches, multi-resolution ring buffers, and random convolutional kernel transform models, that integrate host behavior and network context to enable continuous anomaly fingerprinting and rapid swarm vaccinations.
 - Designed lightweight, collaborative IDS architectures, using Bloom filters with distributed ensemble LSTM Autoencoders, to optimize autonomous placement and scheduling under resource constraints and majority-dishonest collaborator scenarios.
 - Implemented autonomous trust-based scheduling algorithms, using Inception Time-based Convolutional Neural Networks, Dilated Temporal CNNs, Ghost Convolutions, Temporally Downsampled Recurrent Neural Networks, Depthwise Separable CNN, Time Series Mixer, and Multi-Layer Perceptron models, for managing heterogeneous IDS distributions.
- 2019 – 2021
- **Master of Technology (M.Tech.): NIT Durgapur**
Focus: Game-Theoretic Frameworks & Smart City Optimization
 - Formulated game-theoretic models and enhanced incentive schemes, using contribution-based exploration and exploitation models, to optimize volunteer mobility and surplus food distribution in dynamic urban environments.
 - Devised dynamic preference aggregation mechanisms, using bucketed Shapley Scores and K-Means clustering, to optimize crowdsourced project selection and enterprise decision-making for tourist spots.
 - Contributed to intelligent urban transportation by optimizing curvelet entropy features for voice-assisted traffic sign recognition.

RESEARCH PUBLICATIONS

PUBLISHED AND ACCEPTED JOURNAL ARTICLES

1. **Sanyal, S.**, Khatua, M., & Chattopadhyay, P. (2026). CIPHER: A Collaborative IDS Placement and Scheduling Framework for Host-Oriented Mimicry Attack Detection in IoT Networks. *IEEE Internet of Things Journal*. (IF: 8.9). [doi:10.1109/JIOT.2026.3689273](https://doi.org/10.1109/JIOT.2026.3689273)
2. **Sanyal, S.**, Khatua, M., & Chattopadhyay, P. (2026). LITMUS: A Lightweight Collaborative Intrusion Detection System for Host Oriented Mimicry Attack Detection in Industrial Internet of Things Networks with Dishonest Collaborators in the Majority. *Computer Networks*, 286, 112474–112492. (IF: 4.7). [doi:10.1016/j.comnet.2026.112474](https://doi.org/10.1016/j.comnet.2026.112474)
3. **Sanyal, S.**, Mukhopadhyay, S., & Xhafa, F. (2023). Enhanced Incentive Schemes for Volunteers to Distribute Surplus Food in Smart Cities. *Sustainable Cities and Society*, 91, 104409. (IF: 13.3). [doi:10.1016/j.scs.2023.104409](https://doi.org/10.1016/j.scs.2023.104409)
4. **Sanyal, S.**, Kumar Singh, V., Xhafa, F., Sanyal, B., & Mukhopadhyay, S. (2021). A Game Theoretic Framework for Surplus Food Distribution in Smart Cities and Beyond. *Applied Sciences*, 11(11), 5058. (IF: 2.9). [doi:10.3390/app11115058](https://doi.org/10.3390/app11115058)

JOURNAL ARTICLES UNDER REVIEW AND REVISION

1. **Sanyal, S.**, Khatua, M., & Chattopadhyay, P. (2026). MIRAGE: An Adaptive Open-Set Spatio-Temporal Graph Diffusion Ensemble for Explainable Zero-Day Mimicry Detection at the IoT Edge. *Expert Systems With Applications*, 1–42.
2. **Sanyal, S.**, Khatua, M., & Chattopadhyay, P. (2026). HADRON: Autonomous Placement and Scheduling of Heterogeneous IDS using Silent Trust for Mimicry Attack Detection in IoT Networks. *Computer Networks*, 1–19.
3. **Sanyal, S.**, Khatua, M., & Chattopadhyay, P. (2026). JARVIS: Ultralight Conjoint Mimicry Attack Detection with Continuous Anomaly Fingerprinting and Fast Swarm Vaccinations in IoT Networks. *Internet of Things*, 1–24.
4. **Sanyal, S.**, Mukhopadhyay, S., Sanyal, B., & Mukhopadhyay, J. (2026). A Dynamic Preference Aggregation Mechanism using Shapley Scores and K-Means Clustering for Enterprise Decision Making in Smart Cities. *Knowledge and Information Systems*, 1–26.

5. **Sanyal, S.**, Khatua, M., & Chattopadhyay, P. (2026). SIPHON: A Parent Switch Indifferent Placement and Scheduling of Lightweight IDS for Mimicry Attack Detection in IoT Networks. *IEEE Transactions on Reliability*, 1–12.
6. Kumar, R., **Sanyal, S.**, Mukhopadhyay, S., & Xhafa, F. (2025). Volunteer Mobility Management for Surplus Food Distribution in Dynamic Environments in Smart Cities. *Sustainable Cities and Society*, 1–94. **(Under Revision)**.

CONFERENCE PROCEEDINGS

1. **Sanyal, S.**, Mukhopadhyay, S., Xhafa, F., & Mukhopadhyay, J. (2026, July). A Survey of Project Selection in Tourist Spots using Smart City Platforms. In *Proc. of the 20th International Conference on Complex, Intelligent, and Software Intensive Systems (CISIS-2026)* (pp. 1–12). **(In Press)**. Springer. [doi:10.1007/978-3-032-29430-2_11](https://doi.org/10.1007/978-3-032-29430-2_11)
2. Sanyal, B., & **Sanyal, S.** (2026, June). Traffic Sign Recognition with Voice Assistance Using Optimized Curvelet Entropy Features. In *Proc. of the 3rd International Conference of the IEEE Guwahati Subsection (GCON 2026)* (pp. 1–6). IEEE. [doi:10.1109/GCON69192.2026.11648793](https://doi.org/10.1109/GCON69192.2026.11648793)
3. **Sanyal, S.**, Khatua, M., & Chattopadhyay, P. (2026, June). CSIDS: Detecting Mimicry Attacks using Network Context with Host Behavior in IoT Networks. In *Proc. of the 3rd International Conference of the IEEE Guwahati Subsection (GCON 2026)* (pp. 1–6). IEEE. [doi:10.1109/GCON69192.2026.11649178](https://doi.org/10.1109/GCON69192.2026.11649178)
4. **Sanyal, S.**, Priya, T., Debnath, N., Mitra, D., & Mukhopadhyay, S. (2026, June). Placing Tourist Spots in Smart Cities: A Crowdsourced Game Theoretic Approach. In *Proc. of the 3rd International Conference of the IEEE Guwahati Subsection (GCON 2026)* (pp. 1–6). IEEE. [doi:10.1109/GCON69192.2026.11648713](https://doi.org/10.1109/GCON69192.2026.11648713)
5. **Sanyal, S.**, Mukhopadhyay, S., Xhafa, F., & Mukhopadhyay, J. (2026, May). FIREBOLT: Visitor Frequency Aware Crowdsourced Project Selection for Tourist Spots in Smart Cities using Bucketed Shapley Scores. In *Proc. of the 14th International Conference on Emerging Internet, Data & Web Technologies (EIDWT-2026)* (Vol. 291, pp. 255–266). Springer. [doi:10.1007/978-3-032-22798-0_23](https://doi.org/10.1007/978-3-032-22798-0_23)
6. **Sanyal, S.**, Khatua, M., & Chattopadhyay, P. (2024, December). An Adaptive IDS Scheduling Framework for Host Based Mimicry Attack Detection in IoT Networks. In *Proc. of 2024 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS 2024)* (pp. 1–6). IEEE. [doi:10.1109/ANTS63515.2024.10898203](https://doi.org/10.1109/ANTS63515.2024.10898203)
7. **Sanyal, S.**, Mukhopadhyay, S., Xhafa, F., & Mukhopadhyay, J. (2024, June). Crowdsourced Project Selection in Tourist Spots with Voting in Smart Cities. In *Proc. of the Decision Science Alliance International Summer Conference (DSA ISC 2024)* (pp. 105–119). Springer. [doi:10.1007/978-3-031-78241-1_10](https://doi.org/10.1007/978-3-031-78241-1_10)
8. **Sanyal, S.**, Mukhopadhyay, S., Xhafa, F., Singh, V. K., & Sanyal, B. (2022, August). An Incentive Framework for Surplus Food Distribution in the Context of Smart Cities. In *Proc. of the International Conference on Network-Based Information Systems (NBIS 2022)* (pp. 35–47). Springer. [doi:10.1007/978-3-031-14314-4_4](https://doi.org/10.1007/978-3-031-14314-4_4)

BOOKS AND CHAPTERS

1. **Sanyal, S.**, Mukhopadhyay, S., Xhafa, F., Kumar Singh, V., & Sanyal, B. (2023, May). Incentives in Surplus Food Distribution for Smart Cities and Beyond: An Activity Aware Solution. In *Advances in engineering and information science toward smart city and beyond* (pp. 75–90). [doi:10.1007/978-3-031-29301-6_4](https://doi.org/10.1007/978-3-031-29301-6_4)

INDUSTRY EXPERIENCE

2011 – 2019 > **Bio-Statistical Programmer, Data Modeler, and Data Analyst.**
Tata Consultancy Services (TCS), Delhi-NCR, India.

1. I have worked for a Danish Pharmaceutical Client* in the Lifesciences domain as a SAS Bio-statistical programmer doing data analysis and interpretation to establish the safety and efficacy of upcoming drugs undergoing clinical trials in phases 1, 2, and 3.
2. I have also worked for an American Insurance Client* in the Healthcare industry for health-related data modeling and trend prediction of US citizens.

* *Note:* Names of the client organizations have not been disclosed due to contractual obligations.

TEACHING ASSISTANCE & MENTORSHIP EXPERIENCE

TEACHING ASSISTANCE

- IIT Guwahati > **Relational Database Management Systems** for **Coursera** during *September to November 2026, 2025, 2024.*
- IIT Guwahati > **Cloud Computing** for **Coursera** during *September to November 2026.*
- IIT Guwahati > **Database Management Systems** for the **Institute** during *January to May 2026.*
- IIT Guwahati > **Internet of Things** for the **Institute** during *July to November 2024, 2023, 2022.*
- IIT Guwahati > **Computer Systems** for the **Institute** during *January to May 2024.*
- IIT Guwahati > **Computer and Network Security** for the **Institute** during *January to May 2023.*
- IIT Guwahati > **Computer Networks Lab** for the **Institute** during *July to November 2022.*
- IIT Guwahati > **Topics in Networks** for the **Institute** during *January to May 2022.*
- IIT (BHU) Varanasi > **Computer Networks** for the **Institute** during *July to November 2025.*
- IIT (BHU) Varanasi > **Algorithms** for the **Institute** during *January to May 2025.*

MENTORSHIP

- Workshop: 1–Week > **Power Electronics Hardware Design with Microcontroller Programming** at **IIT Guwahati** from *June 24 to June 30, 2023* sponsored by the *Science and Engineering Research Board (SERB), India under the Accelerate Vigyan scheme.*

SKILLS

- Programming > Python, Cython, SLURM, UNIX, Linux, SQL, VBA, C, $\text{\LaTeX}$.
- Tools & Systems > PARAM Shivay and Kamrupa Supercomputers, VS Code, Arduino IDE, MySQL.
- Communication > Reading, writing, speaking: English, Hindi, Bengali, Sanskrit, and Assamese.

MISCELLANEOUS EXPERIENCE

CERTIFICATIONS AND TRAINING

- Workshop: Two–Days > **High Performance Computing (HPC)** organized by the *Indian Institute of Technology Guwahati, India* in collaboration with the *Centre for Development of Advanced Computing (C-DAC), India* on *December 2022.*
- Workshop: Three–Days > **Scaling Blockchains: From Bitcoin to the Lightning Network** organized by the *Indian Institute of Technology Guwahati, India* in collaboration with the *University of Catania, Italy* on *August 2022.*

MISCELLANEOUS EXPERIENCE (continued)

- Sanskrit Linguist > **Diploma Degree** awarded by the *Central Sanskrit University (CSU), Delhi, India*, organized by the *Indian Knowledge Systems (IKS)* in association with the *Indian Institute of Technology Guwahati, India*, on *June 2024*.
- Karate Trainer > **Second Dan Black Belt and Registered Trainer** awarded by the *Traditional Shito-Ryu Karate Do Association (India), TSKA*, approved by the *Karate India Organization (KIO)*, affiliated with the *World Karate Federation (WKF)*, recognized by the *International Olympic Committee (IOC)*, on *December 2022*.

AWARDS AND ACHIEVEMENTS

- Active Participation > **Several competitive and charitable track and field events** from 2011 to 2020.
- Service and Commitment > **TCS Delhi-NCR region** on 2019, 2016, and 2014.
- Outstanding Contribution > **TCS Delhi-NCR region** on 2017, 2016, and 2015.
- Distinguished Training > **TCS Delhi-NCR region** on 2015, 2014, and 2013.
- Best Team of the Year > **TCS Delhi-NCR region** on 2011.

REFERENCES

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Indian Institute of Technology Guwahati,
Guwahati, Assam, India.

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